

Year 1

AERO40005 - Materials 1

PROGRESS TEST n. 2 of 2

To solve in 27 minutes

81%

Section A

Question 1

IMPORTANT INFORMATION:

- Please submit a clear summary of all your answers to Question 1 as the first page of your exam script.
- For multiple choice questions you are required to submit your full written solution along with your chosen answer. Failure to do so, or submission of a wrong solution, will result in a mark of 0. The markers will check your procedure only for questions answered correctly.
- For the true or false questionnaire you are not required to justify your answers. Note however that wrong answers will result in a negative mark, as detailed below.

Part (a)

A metallic plate is subject, in service, to a uniaxial tensile stress $\sigma = 100 \text{ MPa}$. A small central crack of length $2a = 4 \text{ mm}$ is discovered in the plate after an inspection. Assuming $Y = 1$ and knowing that the material has modulus $E = 100 \text{ GPa}$, determine the minimum fracture energy G_{Ic} that the material should have to remain in operation, and state if this should be

- A. $G_{Ic} = 124.3 \text{ J/m}^2$
- (B) $G_{Ic} = 628.3 \text{ J/m}^2$**
- C. $G_{Ic} = 124.3 \text{ kJ/m}^2$
- D. None of the above.

$$G_{Ic} = \frac{K_{Ic}^2}{E} = \frac{(\gamma \sqrt{\pi a})^2}{E} = 200\pi \text{ J/m}^2 = 628.3 \text{ J/m}^2$$

20 [20%]

Part (b)

A unidirectional composite is manufactured by mixing high performance fibres (of modulus $E_F = 100 \text{ GPa}$, density $\rho_F = 2000 \text{ kg/m}^3$ and tensile strength $\sigma_F = 1 \text{ GPa}$) with a polymer matrix ($E_M = 2 \text{ GPa}$, $\rho_M = 1500 \text{ kg/m}^3$), leading to a

material with 10% voids content and fibre volume fraction $\phi_F = 0.4$. Calculate the specific stiffness in the fibre direction E_1 / ρ and state if this is

☒ A. 0.02645 GPa/(kg/m³)

B. 0.1245 GPa/(kg/m³).

C. 0.8657 GPa/(kg/m³)

D. None of the above.

$$\phi_M = 1 - \phi_F - 0.1 = 0.5$$

$$E_1 = \phi_M E_M + \phi_F E_F = 41 \text{ GPa}$$

$$\rho = \phi_M \rho_M + \phi_F \rho_F = 1550 \text{ kg/m}^3$$

$$\therefore \frac{E_1}{\rho} = 0.02645 \text{ GPa/(kg/m}^3\text{)}$$

19 [19%]

Estimate the tensile ductility of this material in the fibre direction, assuming that failure occurs when the stress in the fibres attains the failure stress σ_F , and state if this ductility is

A. $\varepsilon_F = 0.004$

B. $\varepsilon_F = 0.008$

☒ C. $\varepsilon_F = 0.01$

D. None of the above.

$$\varepsilon = \frac{\sigma_F}{E} = 0.01$$

19 [19%]

Part (c)

State if the following are true or false. You will receive 7 points for each correct answer, -7 points for each incorrect answer, 2 points for each unanswered question.

F i. Fracture toughness K_{Ic} is equal to twice the surface tension for brittle materials.

F ii. The degree of crystallinity of a polymer quantifies the average length of its molecular chains.

T iii. Creep deformation of a solid refers to an increase of strain in time while stresses are kept constant in time.

T iv. Ceramics may display ductile behaviour in presence of high hydrostatic compressive stresses.

F v. Increasing the porosity of a solid material decreases its strength, stiffness and densification strain.

T vi.

The modulus of a composite with a continuous matrix and spherical filler particles is comprised between the predictions of inverse and direct rule-of-mixture expressions.

23 [42%]